

SIERTEK Interactive 1: SLM SETUP

Instructor led demonstration using interactive Siertek model

Student Steps:

1. Press On/Off to power up the SLM.
2. Check battery icon If low, replace batteries.
3. Set Range Capacity to 140 dB:
 - Setup, Sig Input, Range Cap, 140 dB, Enter, press Esc.
4. Configure Meter Parameters:
 - Meter 1: Frequency Weighting A; Time Response Slow
5. Set these limits for meter:
 - Threshold: 80 dB
 - Exchange Rate: 3 dB
 - Criterion Level: 85 dB
 - Upper Limit: 115 dB
 - Time: 8 hours

SIERTEK interactive 2: Fast response vs slow response

Hoping to have SLM Interactive that demonstrates how students can measure slow and fast response.

- Steady hum (fan/engine)
- Intermittent bursts (hammering, claps)

The SLM can listen in two ways, *slow* and *fast*. Let's see what that means.

Step 1: Slow Response

- Play the steady sound (hum or fan)
- SLM should be set on A-weighting, Slow.

The numbers/readings should move slowly and stay stable, to show our average which is what a worker would experience over time.

Step 2: Fast Response

- SLM should switch to Z-weighting, Fast.
- Play the burst sound (Intermittent bursts/ hammering or impact tools)

The numbers/ readings should climb quickly to catch sudden spikes/bursts.

SIERTEK Interactive 3: Pre & Post Calibration

ACTIVITY #1: Pre-calibration

1. Press Cal button.
2. Verify last calibration date/time.
3. Press enter button to go into calibration mode
4. Set calibrator to 1000 kHz, 114 dB and attach to mic
5. Let reading stabilize (10 sec)
6. Adjust using UP/DOWN buttons until reading = 114 dB
7. Press Enter to log

ACTIVITY #2: Post-Calibration

1. Press Cal button.
2. Set calibrator to 1000 kHz, 114 dB and attach to mic.
3. You do not have to set the meter to 114 for post calibration, the meter should automatically show 114 (+/- 0.5%)

SIERTEK Interactive 4: Recording documentation

1. From Start Menu "View Current Study."
2. Press first button (eyeglass symbol).
3. Select L_Avg (Meter 1).
4. Record: L_Avg

SIERTEK Interactive 5: Performing Octave band analysis

1. Select a Type I sound level meter with octave band analyzer.
2. The octave band meter should be set at Z (weighting), Fast, and 1/1. This is called a dB all pass (flat or linear reading). Record the findings at different frequencies.
3. Octave bands of concern are 125, 250, 500, 1000, 2000, 4000, and 8000.
4. The tester should hold the meter at the worker's ear level near the equipment (just like a noise source survey). Take measurements for 20 to 30 seconds.
5. Record results for each required octave band (listed above).
6. These results can then be used to calculate a Noise Reduction Rating for hearing protection.

Break down: 1. Turn On the SLM

- Press Power Button

2. Select the Measurement Mode

- Select Measurement Type: From left to right you have buttons that you will press.
|SLM|CAL|FILE|LOCK

The button under SLM will toggle from |SLM|1/1|1/3|

3. Select Frequency Weighting

The octave band meter should be set at Z (weighting) Fast, and 1/1. is called a dB all pass (flat or linear reading). The four buttons are used as toggles to cycle through to set meter.

4. Set Octave Band

- Press the "Octave" Button to choose between different octave band settings.
 - The device will typically display 1/1 Octave or 1/3 Octave. For general surveys, 1/1 Octave is commonly used.

If you're looking for more detailed analysis, switch to 1/3 Octave, which gives narrower frequency bands.

6. Pre-Calibration

- Press the “Calibrate” Button to perform a calibration check:
 - Attach the calibration microphone.
 - Use a calibration device to generate a known sound level
 - Follow the instructions on the display to adjust for proper calibration.

7. Start the Measurement

- Place the Microphone at the desired measurement location, ensuring it's positioned at ear level

Press the “Run/Play” Button to begin recording the noise levels. The Sound Level Pro will now display the noise levels in each of the octave bands (31.5 Hz, 63 Hz, 125 Hz)

You can use the left and right arrows to toggle through the octave bands displayed.

8. Monitor the Results

- The display will show the levels in each frequency band (in dB SPL) as they are measured.
 - The device will show something like typically a graph that you can use the arrows to toggle through:
 - 125 Hz: X dB
 - 250 Hz: X dB
 - 500 Hz: X dB
- Press the “Hold” Button if you want to freeze the display for easier reading of the results.
- Press the “Save” Button to store the data if you need to keep a record of the measurements.

dB(Z)	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz

9. Stop the Measurement

- After you've collected enough data, press the “Stop” Button to end the measurement session.

10. Review & Analyze the Data

- Review the measurements shown on the display.
 - If your device has memory storage, you can review saved data by navigating through the menus.
 - Record or graph the octave band levels (x-axis: frequency, y-axis: dB SPL).

If the Sound Level Pro has a USB or Bluetooth option, you may be able to export the data to a computer for further analysis.

11. Perform Post-Measurement Calibration Check